NOTES ON

COMMUTATIVE ALGEBRA

Lecture Notes & Problem Discussions

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Last updated

August 28, 2026

Abstract

These are lecture notes for a first course in Commutative Algebra, covering rings, ideals, and modules, together with the accompanying problem-set discussions worked out alongside the lectures. The standard references for this material are [AM69], [Bly90], and [DF04].

Contents

Chapter 1

Lecture Notes

1.1 Lecture 1

Convention. Rings are commutative with 1.

Definition 1.1. Let R, S be rings. A *morphism* φ from R to S is a map $\varphi : R \to S$ such that

(1) $\varphi(r+s) = \varphi(r) + \varphi(s)$,

(2) $\varphi(rs) = \varphi(r)\varphi(s)$,

(3) $\varphi(1) = 1$.

Convention. A subring R must contain 1.

Question. What is $R[X]$?

X is not an element of R. Consider

$$S = \{f : \mathbb{N} \to R \mid f(r) = 0 \text{ for all but finitely many } r\},$$

with the usual addition and multiplication in sequences (Cauchy product). Then S is a ring with identity

$$1_S = (1, 0, 0, \ldots, 0).$$

Define $\varphi : R \to S$ by $\varphi(a) = (a, 0, \ldots, 0)$. Then φ is a ring homomorphism, for sure.

Define $X = (0, 1, 0, \ldots)$ and observe $X^2 = (0, 0, 1, 0, \ldots)$. A polynomial P over R can then be identified as an element of S, say

$$P = (P_0, P_1, \ldots, P_n, 0, \ldots) = \varphi(P_0) \cdot 1_S + \varphi(P_1) \cdot X + \cdots + \varphi(P_n) \cdot X^n.$$

Definition 1.2. Let R be a ring. A non-zero, non-unit element $p \in R$ is *prime* if

$$p \mid ab \implies p \mid a \text{ or } p \mid b$$

(include the elementwise definition: for $a, b \in R$, $a \mid b$ if $\exists\, r \in R$ such that $b = r \cdot a$).

Definition 1.3. Let R be a ring. A proper ideal P of R is called a *prime ideal* if

$$ab \in P \implies a \in P \text{ or } b \in P.$$

Definition 1.4. Let R be a ring. $I \subseteq R$ is called an *ideal* if there exists a ring S and a morphism $\varphi : R \to S$ such that $I = \ker(\varphi)$.

Remark 1.5. This immediately gives the elementwise definition of ideals.

$$\mathcal{N}(R) = \{\text{all nilpotent elements of } R\} \quad \rightsquigarrow \quad \text{the } \textit{nil-radical} \text{ of } R.$$

Theorem 1.6.

$$\mathcal{N}(R) = \bigcap_{P \in \operatorname{Spec} R} P, \qquad \text{where } \operatorname{Spec} R = \text{all prime ideals of } R.$$

Proof. Let $x \in \mathcal{N}(R)$. Then for any $P \in \operatorname{Spec} R$, there exists $n_P \in \mathbb{N}$ such that

$$x^{n_P} = 0 \implies 0 \in P \implies x^{n_P} \in P \implies x \in P.$$

Now let $x \notin \mathcal{N}(R)$, and define

$$S = \{I \subseteq R \text{ ideal} \mid x^n \notin I \text{ for all } n\}.$$

$S \neq \emptyset$, as $\langle 0 \rangle \in S$, and S is a poset (ordered by inclusion). By Zorn's Lemma, let M be a maximal element of S.

Claim. M is a prime ideal.

Proof of claim. Let $a \notin M$, $b \notin M$. Now,

$$M + \langle a \rangle,\ M + \langle b \rangle \notin S \implies \exists\, m, n \in \mathbb{N} \text{ such that}$$

$$x^m \in M + \langle a \rangle \text{ and } x^n \in M + \langle b \rangle \implies x^{m+n} \in M + \langle ab \rangle.$$

If $ab \in M$, then $x^{m+n} \in M$, a contradiction. Hence $ab \notin M$, so M is prime. $\square$

Since $x \notin M$ and $M \in \operatorname{Spec} R$, we conclude $x \notin \bigcap_{P \in \operatorname{Spec} R} P$, completing the proof. $\square$

Question. What are the units of $R[X]$?

If R is an integral domain, then they are precisely the units of R.

Proposition 1.7. $f(X) = a_nX^n + \cdots + a_1X + a_0$ is a unit if a_0 is a unit and the a_i's are nilpotent.

Proof. Let $P \in \operatorname{Spec} R$. We have a natural morphism $\varphi : R[X] \to (R/P)[X]$ given by

$$\varphi(b_mX^m + \cdots + b_1X + b_0) = \overline{b_m}X^m + \cdots + \overline{b_1}X + \overline{b_0}, \qquad (\overline{b_i} = b_i \bmod P).$$

Then $\varphi(f)$ is a unit in $(R/P)[X]$, and since R/P is an integral domain, this forces

$$\overline{a_i} \equiv \overline{0} \pmod P \text{ for all } i \in [n], \qquad \text{and} \qquad \overline{a_0} \text{ is a unit of } R/P.$$

That is, $a_0 \notin P$ and $a_1, \ldots, a_n \in P$. As $P \in \operatorname{Spec} R$ was arbitrary,

$$a_0 \notin P \text{ and } a_i \in P \ \ \forall i \in [n], \qquad \text{for all } P \in \operatorname{Spec} R.$$

The other direction is trivial. □

1.2 Lecture 2

Definition 1.8. Let R be a ring and let $(M, +)$ be an abelian group. A *scalar multiplication*

$$R \times M \longrightarrow M, \qquad (r, m) \longmapsto rm,$$

makes M into an R-*module* if the following hold for all $r, s \in R$ and $m, m_1, m_2 \in M$:

(i) $(r + s)(m) = rm + sm$,

(ii) $r(m_1 + m_2) = rm_1 + rm_2$,

(iii) $r(sm) = rsm = s(rm)$,

(iv) $1 \cdot m = m$.

Definition 1.9. If M, N are R-modules, a map $\varphi : M \to N$ is called an R-*module morphism* if

(i) $\varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2)$,

(ii) $\varphi(rm) = r\varphi(m)$.

Definition 1.10. Let R be a ring and S a nonempty set. A *free R-module on S* is a pair (F, f), where F is an R-module and $f : S \to F$ is a set-theoretic map, such that: for any R-module M and any set-theoretic map $g : S \to M$, there exists a unique R-module morphism $h : F \to M$

such that

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f"'] & M \\ F \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

commutes, i.e. $h \circ f = g$.

Proposition 1.11. Let (F, f) be a free R-module on S. Then:

(1) f is injective;

(2) $\mathrm{Im}(f)$ generates F as an R-module.

Proof of (1). Let $x, y \in S$ with $x \neq y$. **Goal:** $f(x) \neq f(y)$.

Take any R-module M with at least two elements, and define $g : S \to M$ such that $g(x) \neq g(y)$. By definition, there exists a (unique) morphism $h : F \to M$ such that $h \circ f = g$. Then

$$(h \circ f)(x) = g(x) \neq g(y) = (h \circ f)(y) \implies f(x) \neq f(y). \qquad \square$$

Definition 1.12. Let M be an R-module and $\emptyset \neq A \subseteq M$. The *submodule of M generated by A* is, equivalently:

(1) the smallest submodule of M containing A;

(2) the intersection of all submodules of M containing A;

(3) the set of all finite R-linear combinations of elements of A.

Proof of (2). Let A be the submodule of F generated by $\mathrm{Im}(f)$. Since $f(s) \in \mathrm{Im}(f) \subseteq A$ for every $s \in S$, f factors as $f = i_A \circ f^*$ for a set-theoretic map $f^* : S \to A$ and the inclusion $i_A : A \hookrightarrow F$:

$$\begin{tikzcd} S \arrow[r, "f^*"] \arrow[dr, "f"'] & A \arrow[d, hook, "i_A"] \\ & F \end{tikzcd}$$

As (F, f) is free on S and A is an R-module, there is a unique morphism $h : F \to A$ such that $h \circ f = f^*$:

$$\begin{tikzcd} S \arrow[r, "f^*"] \arrow[d, "f"'] & A \\ F \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

It is enough to show i_A is onto. We compute

$$(i_A \circ h) \circ f = i_A \circ (h \circ f) = i_A \circ f^* = f.$$

So $i_A \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness in the definition of freeness (applied with $M = F$, $g = f$), $i_A \circ h = \mathrm{id}_F$. Hence i_A is onto, i.e. $A = F$, so $\mathrm{Im}(f)$ generates F. □

Theorem 1.13: Uniqueness of free modules. Let (F, f) be a free R-module on a set S. Then (F', f') is also a free R-module on S if and only if there exists a unique R-module isomorphism $j : F \to F'$ such that $j \circ f = f'$.

Proof. Suppose (F', f') is also free on S. Since (F, f) is free (applied to $M = F'$, $g = f'$), there is a morphism $h : F \to F'$ with $h \circ f = f'$:

$$\begin{tikzcd} S \arrow[r, "f'"] \arrow[d, "f"'] & F' \\ F \arrow[ur, "h"'] & \end{tikzcd}$$

Likewise, since (F', f') is free (applied to $M = F$, $g = f$), there is a morphism $h' : F' \to F$ with $h' \circ f' = f$:

$$\begin{tikzcd} S \arrow[r, "f"] \arrow[d, "f'"'] & F \\ F' \arrow[ur, "h'"'] & \end{tikzcd}$$

Then $(h' \circ h) \circ f = h' \circ (h \circ f) = h' \circ f' = f$. So $h' \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness (with $M = F$, $g = f$), $h' \circ h = \mathrm{id}_F$. Similarly $h \circ h' = \mathrm{id}_{F'}$. So $j := h$ is an isomorphism with $j \circ f = f'$.

Converse. Suppose there is an isomorphism $j : F \to F'$ with $j \circ f = f'$. To show (F', f') is free on S, take any R-module M and set-theoretic map $g : S \to M$; we must produce a unique $h' : F' \to M$ with $h' \circ f' = g$:

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f'"'] & M \\ F' \arrow[ur, dashed, "h'"'] & \end{tikzcd}$$

Since (F, f) is free, there is a unique $h : F \to M$ with $h \circ f = g$. From $j \circ f = f'$ we get $j^{-1} \circ f' = f$, so

$$(h \circ j^{-1}) \circ f' = h \circ (j^{-1} \circ f') = h \circ f = g,$$

i.e. $h' := h \circ j^{-1}$ works:

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f'"'] \arrow[dr, "f"'] & M \\ F' \arrow[r, "j^{-1}"'] & F \arrow[u, "h"'] \end{tikzcd}$$

Uniqueness. Suppose $t : F' \to M$ also satisfies $t \circ f' = g$. Then $t \circ j : F \to M$ satisfies $(t \circ$

$j) \circ f = t \circ (j \circ f) = t \circ f' = g$, so by uniqueness of h (from freeness of (F, f)), $t \circ j = h$, hence $t = h \circ j^{-1} = h'$. □

Theorem 1.14: Existence of free R-modules. For any ring R and nonempty set S, a free R-module on S exists.

Proof. Define

$$F = \{\theta : S \to R \mid \theta(s) = 0 \text{ for almost all } s \in S\}.$$

For $\theta_1, \theta_2 \in F$ and $\lambda \in R$, define

$$(\theta_1 + \theta_2)(s) = \theta_1(s) + \theta_2(s), \qquad (\lambda\theta)(s) = \lambda\,\theta(s).$$

One checks that F is an R-module. Define $f : S \to F$ by $f(s) = \delta_s$, where $\delta_s(t) = 1$ if $t = s$ and 0 otherwise.

Existence of h. Let M be an R-module and $g : S \to M$ a set-theoretic map. Define $h : F \to M$ by

$$h(\theta) = \sum_{s \in S} \theta(s)\, g(s),$$

a finite sum for every $\theta \in F$. For any $t \in S$,

$$(h \circ f)(t) = h(f(t)) = h(\delta_t) = \delta_t(t)\, g(t) = g(t),$$

so $h \circ f = g$:

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f"'] & M \\ F \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

Uniqueness of h. For $\theta \in F$ and $t \in S$,

$$\theta(t) = \theta(t) \cdot 1_R = \sum_{s \in S} \theta(s)\, \delta_s(t) = \left(\sum_{s \in S} \theta(s)\, \delta_s\right)(t) \qquad \forall\, t \in S,$$

so $\theta = \sum_{s \in S} \theta(s)\, \delta_s$. Now let $h' : F \to M$ be any R-module morphism with $h' \circ f = g$. Then for $\theta \in F$,

$$h'(\theta) = h'\left(\sum_{s \in S} \theta(s)\, \delta_s\right) = \sum_{s \in S} \theta(s)\, h'(\delta_s) = \sum_{s \in S} \theta(s)\, h'(f(s)) = \sum_{s \in S} \theta(s)\, g(s) = h(\theta).$$

Hence $h = h'$. □

1.3 Lecture 3

Theorem 1.15. Let R be a ring such that every prime ideal is principal. Then every ideal is principal.

Proof. Let $S = \{\text{all non-principal ideals of } R\}$. For the sake of contradiction (FTSOC), suppose $S \neq \emptyset$. By Zorn's Lemma, let I be a maximal element of S.

Since I is not principal, I is not prime by hypothesis, so there exist $a, b \in R$ with $a \notin I$, $b \notin I$, but $ab \in I$.

Let $J = \langle a \rangle + I \supsetneq I$; since I is maximal among non-principal ideals, $J \notin S$, so J is principal, say $J = \langle c \rangle$.

Also let $K = \{r \in R \mid ra \in I\} \supsetneq I$ (since $b \in K \setminus I$); again $K \notin S$, so $K = \langle d \rangle$ for some d.

$I \subseteq \langle cd \rangle$. Let $x \in I$. Since $I \subseteq J = \langle c \rangle$, there is $\lambda \in R$ with $\lambda c = x$. Now $\lambda a \in I$, so $\lambda \in K = \langle d \rangle$, and hence $x \in \langle cd \rangle$. Thus $I \subseteq \langle cd \rangle$.

$\langle cd \rangle \subseteq I$. There exist $u \in R$, $v \in I$ such that $c = ua + v$ (since c generates $J = \langle a \rangle + I$). Also $da \in I$ (since d generates K). Then

$$cd = (ua + v)d = u(ad) + vd \in I,$$

since $ad = da \in I$ and $v \in I$. Hence $\langle cd \rangle \subseteq I$.

Together, $I = \langle cd \rangle$, contradicting $I \in S$. So $S = \emptyset$, i.e. every ideal of R is principal. □

Theorem 1.16. Let R be a ring such that every prime ideal is finitely generated. Then every ideal of R is finitely generated.

Proof. Let $A = \{I \text{ ideal of } R \mid I \text{ is not finitely generated}\}$. By Zorn's Lemma, let M be a maximal element of A.

Claim: M is a maximal ideal. Suppose not, so there is an ideal N with $M \subsetneq N \subsetneq R$. Since N is maximal (hence prime), it is finitely generated by hypothesis: there exist $a_1, \ldots, a_n \in R$ such that $N = \langle a_1, \ldots, a_n \rangle$.

Since $M \subsetneq N$, some $a_i \notin M$. Then $M + \langle a_i \rangle \supsetneq M$, so by maximality of M in A, $M + \langle a_i \rangle \notin A$; hence there exist $b_1, \ldots, b_k \in R$ such that

$$M + \langle a_i \rangle = \langle b_1, \ldots, b_k \rangle.$$

□

Lemma 1.17. Let I, J_1, J_2 be ideals of R such that $I \subseteq J_1 \cup J_2$. Then either $I \subseteq J_1$ or $I \subseteq J_2$.

Proof. FTSOC, suppose $I \setminus J_1 \neq \emptyset$ and $I \setminus J_2 \neq \emptyset$. Let $x \in I \setminus J_1$ and $y \in I \setminus J_2$; since $I \subseteq J_1 \cup J_2$, this forces $x \in J_2$ and $y \in J_1$. Also $x + y \in I \subseteq J_1 \cup J_2$, so $x + y \in J_1$ or $x + y \in J_2$; WLOG $x + y \in J_1$. Since $y \in J_1$, $x = (x + y) - y \in J_1$, contradicting $x \notin J_1$. □

Theorem 1.18: Prime Avoidance. Let $I, J_1, \ldots, J_n$ be ideals of R such that

(1) $I \subseteq J_1 \cup \cdots \cup J_n$,

(2) $J_3, \ldots, J_n \in \operatorname{Spec} R$.

Then $I \subseteq J_i$ for some i.

Proof. We induct on n. The case $n = 2$ needs no primality hypothesis and is the lemma above; assume $n \geq 3$ and the result for any collection of $n-1$ such ideals.

We prove the contrapositive: if $I \not\subseteq J_i$ for every i, then $I \not\subseteq \bigcup_{i=1}^n J_i$.

Fix i. Apply the induction hypothesis to I together with the $n-1$ ideals $\{J_j\}_{j \neq i}$ (which still has all but at most two members prime): if $I \subseteq \bigcup_{j \neq i} J_j$, the induction hypothesis would give $I \subseteq J_j$ for some $j \neq i$, contradicting our assumption. So $I \not\subseteq \bigcup_{j \neq i} J_j$, and we may choose

$$z_i \in I \setminus \bigcup_{j \neq i} J_j.$$

Since $I \subseteq \bigcup_{i=1}^n J_i$ by hypothesis (1), and $z_i \notin J_j$ for every $j \neq i$, we must have $z_i \in J_i$, for every i.

Set

$$z := z_1 z_2 \cdots z_{n-1} + z_n \in I$$

(a sum of an element of I and a product of elements of I, hence itself in I). We show $z \notin J_i$ for every i, which gives $I \not\subseteq \bigcup_{i=1}^n J_i$, as desired.

- *If $i \in \{1, \ldots, n-1\}$:* since $z_i \in J_i$ is one of the factors of $z_1 \cdots z_{n-1}$, this product lies in J_i. If $z \in J_i$, then $z_n = z - z_1 \cdots z_{n-1} \in J_i$, contradicting $z_n \notin J_i$ (as $i \neq n$).
- *If $i = n$:* if $z \in J_n$, then since $z_n \in J_n$ as well, $z_1 \cdots z_{n-1} = z - z_n \in J_n$. As J_n is prime, some factor $z_k \in J_n$ for a $k \in \{1, \ldots, n-1\}$, contradicting $z_k \notin J_n$ (as $k \neq n$).

This completes the induction. □

Proposition 1.19. An element of $\mathbb{Q}(i)$ lies in $\mathbb{Z}[i]$ if and only if it is the root of a monic polynomial over $\mathbb{Z}$.

Proof. Let $a + bi \in \mathbb{Q}(i)$. Then $a + bi$ is a root of

$$P(X) = X^2 - 2aX + (a^2 + b^2) \in \mathbb{Q}[X].$$

Claim: $2a \in \mathbb{Z}$ and $a^2 + b^2 \in \mathbb{Z}$ together imply $a, b \in \mathbb{Z}$.

Since $2a \in \mathbb{Z}$, $(2a)^2 \in \mathbb{Z}$. Since $a^2 + b^2 \in \mathbb{Z}$, $4(a^2 + b^2) = (2a)^2 + (2b)^2 \in \mathbb{Z}$, so $(2b)^2 = 4(a^2 + b^2) - (2a)^2 \in \mathbb{Z}$. As $2b \in \mathbb{Q}$ has integer square, $2b \in \mathbb{Z}$.

Now $(2a)^2 + (2b)^2 = 4(a^2 + b^2) \equiv 0 \pmod 4$. Since the square of any integer is $\equiv 0 \pmod 4$ if it is even, and $\equiv 1 \pmod 4$ if it is odd, the only way for the sum to be $\equiv 0 \pmod 4$ is

$$(2a)^2 \equiv (2b)^2 \equiv 0 \pmod 4,$$

forcing $2a, 2b$ to both be even, i.e. $a, b \in \mathbb{Z}$.

Since $a + bi \in \mathbb{Z}[i]$ exactly when $a, b \in \mathbb{Z}$, and $a + bi$ is a root of a monic polynomial over $\mathbb{Z}$ exactly when $2a, a^2 + b^2 \in \mathbb{Z}$ (the coefficients of $P(X)$), the claim gives the equivalence. □

Definition 1.20. Let $R \hookrightarrow S$ be rings. An element $\alpha \in S$ is called *integral* over R if α is a root of a monic polynomial

$$X^n + a_{n-1}X^{n-1} + \cdots + a_0$$

over R (i.e. with $a_0, \ldots, a_{n-1} \in R$).

1.4 Lecture 4

Definition 1.21. Let R be a subring of S, and let $U \subseteq S$ be a subset. Define

$$R[U] := \text{the smallest subring of } S \text{ containing } R \text{ and } U.$$

Remark 1.22. For $U, V \subseteq S$,

$$R[U][V] = R[U \cup V] = R[V][U].$$

In particular, if $V = \{u_1, \ldots, u_n\}$, then

$$R[u_1, \ldots, u_n] = R[u_1][u_2] \cdots [u_n].$$

Proposition 1.23.

$$R[u] = \{a_0 + a_1 u + \cdots + a_n u^n \mid a_i \in R,\ n \in \mathbb{N}_{\geq 0}\}.$$

Proof. Let $T = \{a_0 + a_1 u + \cdots + a_n u^n \mid a_i \in R,\ n \in \mathbb{N}_{\geq 0}\}$.

T **is a subring of** S**.** It contains 1 (take $n = 0$, $a_0 = 1$). It is closed under addition: given two such expressions, pad the shorter one with zero coefficients to a common degree and add coefficientwise, which stays in R since R is a ring. It is closed under multiplication: expanding a product of two such expressions and collecting powers of u, each resulting coefficient is a finite sum of products of elements of R, hence lies in R.

$R[u] \subseteq T$**.** Since $R \subseteq T$ (take $n = 0$) and $u \in T$ (take $n = 1, a_0 = 0, a_1 = 1$), and T is a subring of S containing R and u, while $R[u]$ is by definition the *smallest* such subring, we get $R[u] \subseteq T$.

$T \subseteq R[u]$**.** Since $R[u]$ is a subring containing R and u, it is closed under addition and multiplication. So for any $a_0, \ldots, a_n \in R \subseteq R[u]$, each $a_i u^i \in R[u]$, and hence $a_0 + a_1 u + \cdots + a_n u^n \in R[u]$.

Combining both containments, $R[u] = T$. $\square$

Example 1.24. The map $\mathbb{Z}[X] \to \mathbb{Z}[i]$, $X \mapsto i$, is a surjective ring morphism with kernel $\langle X^2 + 1 \rangle$.

Proposition 1.25. Let R, S be rings and $\varphi : R \to S$ be a morphism. Let $\alpha \in S$ be an element. Then φ can be uniquely extended to a morphism $\widetilde{\varphi} : R[X] \to S$ such that $\widetilde{\varphi}|_R = \varphi$ and $\widetilde{\varphi}(X) = \alpha$.

Proof sketch. Define $\widetilde{\varphi} : R[X] \to S$ by

$$\widetilde{\varphi}(a_0 + a_1 X + \cdots + a_n X^n) = \varphi(a_0) + \varphi(a_1)\alpha + \cdots + \varphi(a_n)\alpha^n,$$

and $\widetilde{\varphi}(A \cdot B) = \widetilde{\varphi}(A) \cdot \widetilde{\varphi}(B)$. $\square$

Let R be a subring of S, and let $u \in S$. Then there is a unique morphism $\varphi : R[X] \to S$ such that $\varphi|_R = \mathrm{id}_R$ and $\varphi(X) = u$ (apply the above with $\varphi = \mathrm{id}_R$, $\alpha = u$). Then $\mathrm{Im}(\varphi) = R[u]$; letting $I = \ker \varphi$,

$$R[X]/I \cong R[u].$$

Notation. $I \cap R = \langle 0 \rangle$.

Conversely, let I be an ideal of $R[X]$ such that $I \cap R = \langle 0 \rangle$. We have the natural projection map

$$R[X] \longrightarrow R[X]/I.$$

Since $I \cap R = 0$, $R \hookrightarrow R[X]/I$, so we can identify $a \in R$ with its image $a + I$.

Question. What is the image of $R[X]$ in $R[X]/I$? (It will be determined by $\mathrm{Im}(R)$ and that of X.)

Under the projection, $X \mapsto X + I$. Writing $u = X + I$,

$$R[X]/I \cong R[u].$$

Definition 1.26. Let $R \hookrightarrow S$. S is said to be an *integral extension* of R if each $\alpha \in S$ is integral over R.

Notation. (Recall.) Let R be a commutative ring with 1, and let $A \in M_n(R)$. Then

$$A \cdot \mathrm{adj}(A) = \mathrm{adj}(A) \cdot A = \det(A) \cdot I_n.$$

Theorem 1.27. Let $R \hookrightarrow S$ and $b \in S$. Then

$$b \text{ is integral over } R \iff R[b] \text{ is a finitely generated } R\text{-module.}$$

Proof. ($\Rightarrow$) Assume b is integral over R; then there is a monic polynomial $f(X) \in R[X]$ such that

$f(b) = 0$. Let $\deg f = n \geq 1$. Take any $g(b) \in R[b]$. By the division algorithm,

$$g(X) = q(X)f(X) + r(X), \qquad \deg r < n,$$

so $g(b) = r(b)$. Hence $R[b]$ is generated (as an R-module) by $\{1, b, \dots, b^{n-1}\}$.

($\Leftarrow$) Conversely, let $R[b]$ be a finitely generated R-module, say $R[b] = \langle w_1, \dots, w_n \rangle$. Then for each $i \in [n]$,

$$bw_i = a_{i1}w_1 + \cdots + a_{in}w_n, \qquad a_{ij} \in R,$$

i.e. $(bI - A)\vec{w} = 0$, where $A = (a_{ij})$. Multiplying by $\mathrm{adj}(bI - A)$,

$$\det(bI - A)\, w_i = 0 \qquad \forall\, i \in [n].$$

Since $1 \in R[b] = \langle w_1, \dots, w_n \rangle$, we get $\det(bI - A) = 0$, which is a monic equation in b (of degree n). Hence b is integral over R. □

Theorem 1.28. Let $R \hookrightarrow S$, and let $b_1, b_2, \dots, b_n \in S$. Then each b_i is integral over R if and only if $R[b_1, \dots, b_n]$ is a finitely generated R-module.

Proof. Induct on n. The case $n = 1$ is the previous theorem.

For $n > 1$: assume $b_1, \dots, b_n$ are integral over R. In particular b_n is integral over R, hence also integral over $R[b_1, \dots, b_{n-1}]$; by the previous theorem (applied over the base ring $R[b_1, \dots, b_{n-1}]$), $R[b_1, \dots, b_{n-1}][b_n]$ is a finitely generated $R[b_1, \dots, b_{n-1}]$-module, which (combined with the induction hypothesis that $R[b_1, \dots, b_{n-1}]$ is a finitely generated R-module) makes $R[b_1, \dots, b_n]$ a finitely generated R-module.

The converse is the same as the $n = 1$ case. □

Exercise 1.29. Let $R \hookrightarrow S$, and let $a, b \in S$ be integral over R. Then $a \pm b$ and ab are integral over R.

1.5 Lecture 5

Corollary 1.30. Let $R \hookrightarrow S$ and $b_1, \dots, b_n \in S$ be integral over R (equivalently, $R[b_1, \dots, b_n]$ is a finitely generated R-module). Then every $b \in R[b_1, \dots, b_n]$ is integral over R.

Proof. Since $b \in R[b_1, \dots, b_n]$,

$$R[b_1, \dots, b_n] = R[b_1, \dots, b_n, b],$$

which is a finitely generated R-module. By the previous theorem, b is integral over R. □

Proposition 1.31. Let $A \subseteq B \subseteq C$ be rings such that B is integral over A and C is integral over B. Then C is integral over A.

Proof. Let $c \in C$. Then c is integral over B:

$$c^n + b_{n-1}c^{n-1} + \cdots + b_0 = 0, \qquad b_i \in B.$$

Let $R := A[b_0, \ldots, b_{n-1}]$. Then c is integral over R, so $R[c]$ is a finitely generated R-module.

Since $b_0, \ldots, b_{n-1}$ are integral over A, R is a finitely generated A-module. Hence $R[c]$ is a finitely generated A-module, and since $c \in R[c]$, c is integral over A. □

Definition 1.32. Define $\overline{R}^S := \{r \in S \mid r \text{ is integral over } R\}$. Note that $\overline{R}^S$ is a subring of S. This $\overline{R}^S$ is called the *integral closure* of R in S.

Definition 1.33. R is said to be *integrally closed in* S if $\overline{R}^S = R$.

$$R \hookrightarrow \overline{R}^S \hookrightarrow S.$$

Observation 1.34. $\overline{R}^S$ is integrally closed in S.

Definition 1.35. Let R be an integral domain with quotient field $F = \mathbb{Q}(R)$. We say that R is *integrally closed* if R is integrally closed in F.

Proposition 1.36. If R is a UFD, then R is integrally closed.

Proof. Let $F = \mathbb{Q}(R)$. Say $r/s \in F$ with $r, s \in R$, $s \neq 0$, is integral over R, and suppose r, s have no common prime factors. Then

$$\left(\frac{r}{s}\right)^n + a_{n-1}\left(\frac{r}{s}\right)^{n-1} + \cdots + a_1\left(\frac{r}{s}\right) + a_0 = 0, \qquad a_i \in R,$$

so

$$r^n + a_{n-1}r^{n-1}s + \cdots + a_1 r s^{n-1} + a_0 s^n = 0.$$

Since s divides every term but the first, $s \mid r^n$. FTSOC, if s is not a unit, it has a prime factor p; then $p \mid r^n$, and since p is prime, $p \mid r$ – contradicting that r, s have no common prime factors. Hence s is a unit in R, so $r/s \in R$. □

Example 1.37. $\mathbb{Z}[\sqrt{-5}]$ is integrally closed but not a UFD. (Exercise.)

Example 1.38. $\mathbb{Z}[\sqrt{5}]$ is an integral domain which is not integrally closed.

Proof. $\mathbb{Q}(\mathbb{Z}[\sqrt{5}]) = \mathbb{Q}(\sqrt{5})$, and
$$\frac{1+\sqrt{5}}{2} \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Z}[\sqrt{5}]$$
is a root of the monic polynomial $X^2 - X - 1 \in \mathbb{Z}[X]$, hence integral over $\mathbb{Z}[\sqrt{5}]$, yet not itself an element of $\mathbb{Z}[\sqrt{5}]$. $\square$

Extensions and Contractions

Let $\varphi : R \to S$ be a ring morphism. Then φ factors as
$$R \xrightarrow{\varphi} \varphi(R) \xhookrightarrow{i} S \qquad \text{with} \qquad R/\operatorname{Ker}\varphi \cong \varphi(R).$$

(1) If $I \subseteq R$ is an ideal, $\varphi(I)$ need *not* be an ideal of S.

(2) If $J \subseteq S$ is an ideal, $\varphi^{-1}(J)$ *is* an ideal of R.

(3) If $Q \in \operatorname{Spec} S$, then $\varphi^{-1}(Q) \in \operatorname{Spec} R$.

Question. Suppose $P \in \operatorname{Spec} R$. Can we say that $\langle \varphi(P) \rangle$ is prime in S?

Consider $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$: $\langle 2 \rangle \mapsto \langle 1+i \rangle \langle 1-i \rangle$. If $p \equiv 1 \pmod 4$ is prime, then $\langle p \rangle$ is *not* prime in $\mathbb{Z}[i]$ (so the answer to the question above is no, in general).

Exercise 1.39. If $p \equiv 3 \pmod 4$, then $\langle p \rangle$ is a prime ideal in $\mathbb{Z}[i]$.

Definition 1.40. Let $\varphi : R \to S$ be a ring morphism.

- For I an ideal of R, the *extension* I^e is the ideal of S generated by $\varphi(I)$.
- For J an ideal of S, the *contraction* $J^c := \varphi^{-1}(J)$, an ideal of R.

Exercise 1.41.
$$I \subseteq I^{ec}, \qquad J \supseteq J^{ce}, \qquad I^e = I^{ece}, \qquad J^c = J^{cec}.$$

Rings of Fractions

Definition 1.42. Let R be a ring. A subset $S \subseteq R$ is called *multiplicative* if

(1) $1 \in S$,

(2) $a, b \in S \implies ab \in S$.

Construction 1.43. On $R \times S$, define the relation $\sim$ by

$$(a,s) \sim (b,t) \iff \exists\, u \in S \text{ such that } u(at - bs) = 0.$$

$\sim$ is transitive. Suppose $(a,s) \sim (b,t)$ and $(b,t) \sim (c,u)$; so $(at-bs)v = 0$ for some $v \in S$, and $(bu - ct)w = 0$ for some $w \in S$. Then

$$atv - bsv = 0 \rightsquigarrow uw{\cdot}atv - bsuw = 0, \qquad buw - ctw = 0 \rightsquigarrow buwsv - ctwsv = 0.$$

Adding,

$$(au - cs)\, vtw = 0, \qquad vtw \in S,$$

so $(a,s) \sim (c,u)$. $\square$

Notation. The equivalence class of (a,s) is denoted $\dfrac{a}{s}$.

Let

$$S^{-1}R := \{\text{all equivalence classes}\} = \left\{ \frac{a}{s} \,\middle|\, a \in R,\ s \in S \setminus \{0\} \right\}.$$

Define

$$\frac{a}{s} + \frac{b}{t} := \frac{at+bs}{st} \quad \text{(well-defined)}, \qquad \frac{a}{s} \cdot \frac{b}{t} := \frac{ab}{st} \quad \text{(well-defined)}.$$

$S^{-1}R$ is a commutative ring with 1. We have a canonical ring morphism

$$\varphi : R \to S^{-1}R, \qquad r \mapsto \frac{r}{1}.$$

Remark 1.44. φ is not injective in general.

Proposition 1.45. Let $g : R \to A$ be a ring morphism such that $g(S) \subseteq A^\times$ (the units of A). Then there is a unique morphism $h : S^{-1}R \to A$ such that

$$\begin{tikzcd} R \arrow[r, "g"] \arrow[d, "\varphi"'] & A \\ S^{-1}R \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

commutes, i.e. $h \circ \varphi = g$.

Proof. **Uniqueness.** Suppose such an h exists. Then

$$h\left(\frac{r}{1}\right) = h(\varphi(r)) = g(r), \qquad h\left(\frac{1}{s}\right) = h\left(\left(\frac{s}{1}\right)^{-1}\right) = h\left(\frac{s}{1}\right)^{-1} = g(s)^{-1},$$

so

$$h\left(\frac{r}{s}\right) = h\left(\frac{r}{1}\right) h\left(\frac{1}{s}\right) = g(r)\, g(s)^{-1},$$

i.e. h is uniquely determined by g.

Existence. Define $h : S^{-1}R \to A$ by $h(r/s) = g(r)g(s)^{-1}$.

Well-definedness. Suppose $r/s = r'/s'$; then there is $t \in S$ with $t(rs' - r's) = 0$, so

$$g(t)\big(g(r)g(s') - g(r')g(s)\big) = 0.$$

Since $g(t) \in A^\times$, $g(r)g(s') = g(r')g(s)$, i.e. $g(r)g(s)^{-1} = g(r')g(s')^{-1}$. $\square$

Example 1.46. Let $P \in \operatorname{Spec} R$ and $S = R \setminus P$. Then S is multiplicative; write $S^{-1}R =: R_P$. Let

$$PR_P := \left\{ \frac{p}{s} \;\middle|\; p \in P,\ s \in R \setminus P \right\},$$

an ideal of R_P. If $b/t \notin PR_P$, then $b \notin P$, so $b \in R \setminus P$, so b/t is a unit in R_P. Hence PR_P is the unique maximal ideal of R_P, so R_P is local.

Example 1.47. Let $f \in R$ and $S = \{1, f, f^2, \ldots\}$. Then $R_f := S^{-1}R$.

Construction 1.48. Let R be a ring, S a multiplicative subset of R, and M an R-module. On $M \times S$, define $\sim$ by

$$(m, s) \sim (m', s') \iff \exists\, t \in S \text{ such that } t(ms' - m's) = 0.$$

Then $S^{-1}M := (M \times S)/\!\sim$ is an $S^{-1}R$-module.

Let M, N be R-modules and $\varphi : M \to N$ an R-module morphism. Define $S^{-1}\varphi : S^{-1}M \to S^{-1}N$ by

$$\frac{m}{s} \longmapsto \frac{\varphi(m)}{s}.$$

$S^{-1}\varphi$ is a well-defined $S^{-1}R$-module morphism.

Proposition 1.49. $S^{-1}(-)$ preserves exactness: if

$$M' \xrightarrow{\varphi} M \xrightarrow{\psi} M''$$

is exact at M, then

$$S^{-1}M' \xrightarrow{S^{-1}\varphi} S^{-1}M \xrightarrow{S^{-1}\psi} S^{-1}M''$$

is exact at $S^{-1}M$.

Proof. We have $\psi \circ \varphi = 0$, so $(S^{-1}\psi) \circ (S^{-1}\varphi) = S^{-1}(\psi \circ \varphi) = 0$. Hence $\operatorname{Im}(S^{-1}\varphi) \subseteq \operatorname{Ker}(S^{-1}\psi)$.

Conversely, let $m/s \in \operatorname{Ker}(S^{-1}\psi)$. Then $\psi(m)/s = 0$, so there is $t \in S$ with $t\,\psi(m) = 0$, i.e.

$\psi(tm) = 0$, i.e. $tm \in \operatorname{Ker}(\psi) = \operatorname{Im}(\varphi)$. So there is $m' \in M'$ with $\varphi(m') = tm$. Then

$$\frac{m}{s} = \frac{mt}{st} = \frac{\varphi(m')}{st} = (S^{-1}\varphi)\left(\frac{m'}{st}\right) \in \operatorname{Im}(S^{-1}\varphi).$$

Hence $\operatorname{Ker}(S^{-1}\psi) \subseteq \operatorname{Im}(S^{-1}\varphi)$, giving equality. □

1.6 Lecture 6

Theorem 1.50. Let M be an R-module generated by n elements, and let

$$\varphi \in \operatorname{Hom}(M, M) := \{f : M \to M \mid f \text{ is an } R\text{-module morphism}\}.$$

Let I be an ideal of R such that $\varphi(M) \subseteq IM$. Then there is a relation of the form

$$\varphi^n + a_{n-1}\varphi^{n-1} + \cdots + a_0 = 0, \qquad a_i \in I^i.$$

Proof. Let $M = Rw_1 + \cdots + Rw_n$, so $IM = Iw_1 + \cdots + Iw_n$; in particular $\varphi(M) \subseteq IM$. Write

$$\varphi(w_i) = \sum_{j=1}^{n} a_{ij} w_j, \qquad a_{ij} \in I \text{ for all } i,$$

i.e.

$$\sum_{j=1}^{n} (\varphi\, \delta_{ij} - a_{ij}) w_j = 0.$$

The coefficient matrix has (i, j)-entry $\varphi\, \delta_{ij} - a_{ij}$, with entries in the commutative subring of $\operatorname{End}(M)$ generated by R (acting by scalars) and φ. Let $d = \det(\text{coefficient matrix})$. Multiplying by the adjugate matrix,

$$d \cdot w_k = 0 \qquad \forall\, k \in [n],$$

so $d \cdot M = 0$, i.e. d is the zero endomorphism of M. Expanding the determinant gives exactly a relation of the stated form. □

Theorem 1.51: Nakayama's Lemma. Let M be a finitely generated R-module and I an ideal of R. If $M = IM$, then there exists $a \in R$ such that $aM = 0$ and $a \equiv 1 \pmod{I}$. Furthermore, if I is the Jacobson radical of R, then $M = 0$.

Proof. Take $\varphi = \mathrm{id}_M$. By the previous theorem, we have

$$1 + a_{n-1} + \cdots + a_0 = 0 \qquad (\text{as endomorphisms of } M).$$

Define $a := 1 + a_{n-1} + \cdots + a_0$. Then $aM = 0$ and $a \equiv 1 \pmod{I}$.

If $I \subseteq J(R)$ (the Jacobson radical), then $a - 1 \in J(R)$, so a is a unit; hence $aM = 0$ forces $M =$

$a^{-1}(aM) = 0$. □

Theorem 1.52. Let M be a finitely generated R-module and I an ideal of R such that $M = IM$. Then there exists $i \in I$ such that $(1+i)M = 0$.

Proof (Ojanguren). Induct on the minimal number of generators $\nu(M) = n$ of M.

If $\nu(M) = 0$, then $M = 0$ and we are done.

Let $M = Rm_1 + \cdots + Rm_n$, and set $M' := M/Rm_n$. Then

$$IM' = \frac{IM + Rm_n}{Rm_n} = \frac{M + Rm_n}{Rm_n} = \frac{M}{Rm_n} = M',$$

using $IM = M$. Since $\nu(M') < n$, by induction there is $x \in I$ such that $(1+x)M' = 0$, i.e.

$$(1+x)\frac{M}{Rm_n} = 0 \qquad \Longrightarrow \qquad (1+x)M \subseteq Rm_n.$$

Then

$$(1+x)M = (1+x)IM = I(1+x)M \subseteq I \cdot Rm_n = Im_n,$$

using $(1+x)M \subseteq Rm_n$ in the last containment. In particular $(1+x)m_n \in Im_n$, so $(1+x)m_n = ym_n$ for some $y \in I$, giving

$$(1+x-y)m_n = 0.$$

Now for any $\mu \in M$, $(1+x)\mu = rm_n$ for some $r \in R$ (since $(1+x)M \subseteq Rm_n$), so

$$(1+x-y)(1+x)\mu = r(1+x-y)m_n = 0.$$

Hence $(1+x)(1+x-y)M = 0$. Expanding, $(1+x)(1+x-y) = 1+i$ where $i := x + (x-y) + x(x-y) \in I$. So $(1+i)M = 0$. □

Definition 1.53. Let M, N be R-modules. The *tensor product* of M, N is a pair (T, θ) such that T is an R-module, $\theta : M \times N \to T$ is bilinear, and for every bilinear map $f : M \times N \to P$, there is a unique linear map $h : T \to P$ such that

$$\begin{tikzcd} M \times N \arrow[r, "f"] \arrow[d, "\theta"'] & P \\ T \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

commutes, i.e. $h \circ \theta = f$.

Recall the canonical morphism $R \to S^{-1}R$, $r \mapsto r/1$, and let I be an ideal of R. An element of the extended ideal I^e is a finite sum

$$\frac{r_1}{s_1} \cdot \frac{x_1}{1} + \cdots + \frac{r_n}{s_n} \cdot \frac{x_n}{1} = \frac{s_1' r_1 x_1 + \cdots + s_n' r_n x_n}{s_1 \cdots s_n},$$

where $r_i \in R$, $s_i \in S$, $x_i \in I$ (writing s_i' for the product of all s_j with $j \neq i$).

Proposition 1.54.

(1) Every ideal of $S^{-1}R$ is an extended ideal.

(2) The prime ideals of $S^{-1}R$ are in one-to-one correspondence, $P \leftrightarrow S^{-1}P$, with the prime ideals P of R such that $S \cap P = \emptyset$.

Proof. See [AM69]. □

Local Properties of Modules

Proposition 1.55. Let M be an R-module. Then the following are equivalent:

(1) $M = 0$;

(2) $M_P = 0$ for all $P \in \text{Spec}(R)$;

(3) $M_\mathfrak{m} = 0$ for all $\mathfrak{m} \in \text{Max}(R)$.

(Here for $P \in \text{Spec}(R)$, writing $S = R \setminus P$, we set $M_P := S^{-1}M$.)

Proof. (1) $\Rightarrow$ (2) $\Rightarrow$ (3) trivially.

(3) $\Rightarrow$ (1): Let $M \neq 0$; then there is $x \in M$ with $x \neq 0$. Let $I = \text{Ann}(x) = \{r \in R \mid rx = 0\}$, a proper ideal of R. Let $\mathfrak{m}$ be a maximal ideal with $I \subseteq \mathfrak{m}$.

Consider $x/1 \in M_\mathfrak{m}$. Since $M_\mathfrak{m} = 0$, $x/1 = 0$, so there is $t \in R \setminus \mathfrak{m}$ with $tx = 0$. Then $t \in I \subseteq \mathfrak{m}$ – contradicting $t \notin \mathfrak{m}$. □

Exercise 1.56. Let $\varphi \in \text{Hom}(M, N)$. TFAE:

(1) φ is injective (resp. surjective);

(2) $\varphi_P : M_P \to N_P$ is injective (resp. surjective) for all $P \in \text{Spec}(R)$;

(3) $\varphi_\mathfrak{m} : M_\mathfrak{m} \to N_\mathfrak{m}$ is injective (resp. surjective) for all $\mathfrak{m} \in \text{Max}(R)$.

Proposition 1.57. Let $R \hookrightarrow S$ be rings with S integral over R.

(1) If J is an ideal of S, then $R/(J \cap R) \hookrightarrow S/J$ is integral.

(2) If T is a multiplicative subset of R, then $T^{-1}R \hookrightarrow T^{-1}S$ is integral.

Proof. (1) is straightforward (reduce the monic integral equation modulo J).

(2) Let $x/t \in T^{-1}S$ ($x \in S$, $t \in T$). Since x is integral over R,

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0 \qquad \text{for some } a_i \in R.$$

Dividing by t^n,

$$\left(\frac{x}{t}\right)^n + \frac{a_1}{t}\left(\frac{x}{t}\right)^{n-1} + \cdots + \frac{a_n}{t^n} = 0,$$

so x/t is integral over $T^{-1}R$. □

Proposition 1.58. Let $R \hookrightarrow S$ be an integral extension, with R, S both integral domains. Then R is a field if and only if S is a field.

Proof. ($\Rightarrow$) Let R be a field and $y \in S \setminus \{0\}$. Since y is integral over R, take a monic relation of minimal degree,

$$y^n + a_1 y^{n-1} + \cdots + a_n = 0, \qquad a_i \in R.$$

If $a_n = 0$, then $y(y^{n-1} + a_1 y^{n-2} + \cdots + a_{n-1}) = 0$; since S is a domain and $y \neq 0$, this gives a relation of degree $n-1$, contradicting minimality. So $a_n \neq 0$, and as R is a field, a_n is a unit. Then

$$y\left(y^{n-1} + \cdots + a_{n-1}\right) = -a_n,$$

and since $-a_n$ is a unit, y is a unit.

($\Leftarrow$) Let S be a field. Take any $x \in R \setminus \{0\}$. Then $x^{-1} \in S$ is integral over R:

$$x^{-m} + a_1 x^{-(m-1)} + \cdots + a_m = 0, \qquad a_i \in R,$$

so

$$x^{-1} = -\left(a_1 + a_2 x + \cdots + a_m x^{m-1}\right) \in R.$$

□

1.7 Lecture 7

Example 1.59. Let k be a field, t an indeterminate, and $A = k[t^2, t^3] \subseteq B = k[t]$. Since A is a subring of the integral domain B, A is an integral domain. The fraction field of both A and B is $k(t)$, and since B is a UFD, B is integrally closed in $k(t)$.

Question. Is A integrally closed in $k(t)$?

No: t is integral over A, being a root of $X^2 - t^2 \in A[X]$, but $t \notin A$. So A is not integrally closed.

Question. What is the integral closure of A in $k(t)$?

It is B, since B is integrally closed in $k(t)$.

Example 1.60. Similarly,

$$\begin{tikzcd}
B \arrow[r, hook] & \mathbb{Q}(i) \\
\mathbb{Z} \arrow[r, hook] \arrow[u, hook] & \mathbb{Q} \arrow[u, hook]
\end{tikzcd}$$

where B is the integral closure of $\mathbb{Z}$ (an integrally closed domain) in $\mathbb{Q}(i)$. One checks $B = \mathbb{Z}[i]$: $\min(i, \mathbb{Q}) = X^2 + 1 \in \mathbb{Z}[X]$.

Theorem 1.61. Let A be an integrally closed domain with $K = \mathrm{Q}(A)$, and let L be an algebraic extension of K. Then $\alpha \in L$ is integral over A if and only if $\min(\alpha, K) \in A[X]$.

Proof. ($\Leftarrow$) is trivial: if $f(X) = X^n + a_1X^{n-1} + \cdots + a_n = \min(\alpha, K) \in A[X]$, then $f(\alpha) = 0$ exhibits α as integral over A.

($\Rightarrow$) Let α be integral over A; we show the coefficients a_i of $f := \min(\alpha, K)$ lie in A.

Let $\overline{L}$ be an algebraic closure of L. In $\overline{L}$, f splits as

$$f(X) = (X - \alpha_1) \cdots (X - \alpha_n),$$

where the α_i are the conjugates of α over K. For each i there is a K-isomorphism $K[\alpha] \cong K[\alpha_i]$ fixing K and sending $\alpha \mapsto \alpha_i$.

Since α is integral over A, there is a monic $g(X) \in A[X]$ with $g(\alpha) = 0$; as $f = \min(\alpha, K)$, $f \mid g$ in $K[X]$. Applying the isomorphism above, $g(\alpha_i) = 0$ for each i, so each α_i is integral over A.

Now $a_1, \ldots, a_n \in A[\alpha_1, \ldots, \alpha_n]$ (they are, up to sign, the elementary symmetric functions of the α_i – Newton's identities), and each α_i is integral over A, so each a_i is integral over A. But $a_i \in K$ and A is integrally closed in K, so $a_i \in A$ for all i. $\square$

Exercise 1.62.

(1) $k[X, Y]/\langle Y^2 - X^3 \rangle \cong k[t^2, t^3]$ (the coordinate ring of the curve $Y^2 = X^3$).

(2) Show directly that $k[t^2, t^3]$ is not a UFD.

Example 1.63. Let A be a UFD with $2 \in A^\times$, and let $f \in A$ be square-free. Then $A[\sqrt{f}]$ is an integrally closed domain.

Proof. Since $\langle X^2 - f \rangle \in \mathrm{Spec}(A[X])$ (as f is square-free, $X^2 - f$ is irreducible),

$$A[\sqrt{f}] \cong A[X]/\langle X^2 - f \rangle$$

is an integral domain.

Let α be a square root of f; we ask whether $A[\alpha]$ is integrally closed. Let $K := \mathrm{Q}(A)$; since A is a

UFD, A is integrally closed in K.

Case $\alpha \in K$: Then $X^2 - f \in A[X]$ has root $\alpha \in K$, so α is integral over A, hence $\alpha \in A$ (as A is integrally closed in K). So $A[\alpha] = A$, trivially integrally closed.

Case $\alpha \notin K$: Then

$$\mathbb{Q}(A[\alpha]) = K(\alpha) = K + K\alpha,$$

and every $\theta \in K(\alpha)$ can be written uniquely as $\theta = x + y\alpha$ with $x, y \in K$; its minimal polynomial over K is

$$X^2 - 2xX + (x^2 - fy^2).$$

Suppose θ is integral over $A[\alpha]$. Since α is integral over A (a root of $X^2 - f \in A[X]$), $A[\alpha]$ is integral over A; by transitivity, θ is integral over A. By the previous theorem (as A is integrally closed in K),

$$X^2 - 2xX + (x^2 - fy^2) \in A[X],$$

i.e. $2x \in A$ and $x^2 - fy^2 \in A$. Since 2 is a unit, $x \in A$; hence $fy^2 = x^2 - (x^2 - fy^2) \in A$.

If some prime $p \in A$ divides the denominator of y (writing $y \in K$ in lowest terms), then since $fy^2 \in A$ and f is square-free, we would need $p^2 \mid f$ – a contradiction. Hence $y \in A$, so $\theta = x + y\alpha \in A[\alpha]$.

Together with the trivial case $A[\alpha] = A$, this shows $A[\alpha]$ is integrally closed. □

Lemma 1.64. Let $A \hookrightarrow B$ be an integral extension. If $P \in \operatorname{Max}(B)$, then $P \cap A \in \operatorname{Max}(A)$. Conversely, if $\mathfrak{p} \in \operatorname{Max}(A)$, then there exists $P \in \operatorname{Spec}(B)$ such that $P \cap A = \mathfrak{p}$ (and P is maximal too).

Proof. $P \cap A \in \operatorname{Spec}(A)$, and $A/(P \cap A) \hookrightarrow B/P$ is an integral extension. Since B/P is a field, $A/(P \cap A)$ is a field, so $P \cap A \in \operatorname{Max}(A)$.

Conversely, let $\mathfrak{p} \in \operatorname{Max}(A)$. It suffices to show $\mathfrak{p}B \neq B$: if $\mathfrak{p}B$ is a proper ideal, there is $P \in \operatorname{Max}(B)$ with $\mathfrak{p}B \subseteq P$, so $\mathfrak{p} \subseteq P \cap A$; since $1 \notin P \cap A$ and $\mathfrak{p}$ is maximal, $P \cap A = \mathfrak{p}$.

Suppose instead $\mathfrak{p}B = B$. Then $1 = b_1 p_1 + \cdots + b_n p_n$ for some $b_i \in B$, $p_i \in \mathfrak{p}$. Let $C = A[b_1, \ldots, b_n]$; since each b_i is integral over A, C is a finitely generated A-module. From $1 = \sum b_i p_i$ (with $b_i \in C$, $p_i \in \mathfrak{p}$), $\mathfrak{p}C = C$.

By Nakayama's Lemma, there is $a \in A$ with $a \equiv 1 \pmod{\mathfrak{p}}$ and $aC = 0$; since $1 \in C$, $a = a \cdot 1 = 0$, so $0 \equiv 1 \pmod{\mathfrak{p}}$, i.e. $1 \in \mathfrak{p}$ – a contradiction. □

Theorem 1.65: Lying Over. Let $A \hookrightarrow B$ be an integral extension and $\mathfrak{p} \in \operatorname{Spec}(A)$.

(i) There exists $P \in \operatorname{Spec}(B)$ such that $P \cap A = \mathfrak{p}$ (we say P *lies over* $\mathfrak{p}$).

(ii) There is no inclusion between distinct prime ideals of B lying over $\mathfrak{p}$: if $P \subseteq Q$ and $P \cap A = \mathfrak{p} = Q \cap A$, then $P = Q$.

Proof. Localize at $S = A \setminus \mathfrak{p}$: we obtain $A_{\mathfrak{p}} \hookrightarrow B_{\mathfrak{p}} := S^{-1}B$, an integral extension (by the localizationintegral-extension theorem below), with $A_\mathfrak{p}$ local with maximal ideal $\mathfrak{p}A_\mathfrak{p}$. By the previous lemma (applied to $A_\mathfrak{p} \hookrightarrow B_\mathfrak{p}$ and the maximal ideal $\mathfrak{p}A_\mathfrak{p}$), there is a maximal ideal $\mathfrak{P}$ of $B_\mathfrak{p}$ with $\mathfrak{P} \cap A_\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}$.

Let $P \subseteq B$ be the contraction of $\mathfrak{P}$ along $B \to B_\mathfrak{p}$. Then $P \cap A = \mathfrak{p}$, proving (i).

Moreover, the primes of B lying over $\mathfrak{p}$ are in bijective correspondence with the maximal ideals of $B_\mathfrak{p}$ lying over $\mathfrak{p}A_\mathfrak{p}$ (via this same contraction/extension correspondence). Since distinct maximal ideals are never comparable by inclusion, neither are the corresponding primes of B lying over $\mathfrak{p}$, proving (ii). □

Theorem 1.66. Let $A \hookrightarrow B$ be an extension of rings and C the integral closure of A in B. If S is a multiplicative subset of A, then $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.

Proof. Let $b/s \in S^{-1}B$ be integral over $S^{-1}A$:

$$\left(\frac{b}{s}\right)^n + \frac{a_1}{s_1}\left(\frac{b}{s}\right)^{n-1} + \cdots + \frac{a_n}{s_n} = 0.$$

Let $t = s_1 \cdots s_n$. Multiplying through by $(st)^n$,

$$(st)^n \left[\left(\frac{b}{s}\right)^n + \cdots\right] = 0$$

becomes a monic equation in bt with coefficients in A. Hence bt is integral over A, i.e. $bt \in C$, so

$$\frac{b}{s} = \frac{bt}{st} \in S^{-1}C.$$

□

1.8 Lecture 8

1.9 Lecture 9

Proposition 1.67. Let $\varphi : A \to B$ be a ring morphism and $\mathfrak{p} \in \operatorname{Spec} A$. Then $\mathfrak{p}$ is the contraction of some $Q \in \operatorname{Spec} B$ if and only if $\mathfrak{p}^{ec} = \mathfrak{p}$.

Proof. In general, $\mathfrak{p} \subseteq \mathfrak{p}^{ec}$.

($\Rightarrow$) If $\mathfrak{p} = Q^c$ for some $Q \in \operatorname{Spec} B$, then

$$\mathfrak{p}^e = Q^{ce} \subseteq Q \quad \Longrightarrow \quad \mathfrak{p}^{ec} \subseteq Q^c = \mathfrak{p}.$$

($\Leftarrow$) Assume $\mathfrak{p}^{ec} = \mathfrak{p}$. Let $S = \varphi(A \setminus \mathfrak{p})$, a multiplicative subset of B.

Note: $\mathfrak{p}^e \cap S = \emptyset$. Indeed, if $x \in \mathfrak{p}^e \cap S$, then $\varphi^{-1}(x) \in A \setminus \mathfrak{p}$; but $x \in \mathfrak{p}^e$ gives $\varphi^{-1}(x) \in \mathfrak{p}^{ec} = \mathfrak{p}$ – a contradiction.

So $\mathfrak{p}^e S^{-1}B$ is a proper ideal of $S^{-1}B$, hence contained in some $\mathfrak{m} \in \operatorname{Max}(S^{-1}B)$. Let Q be the contraction of $\mathfrak{m}$ in B; then $Q \in \operatorname{Spec} B$ and $\mathfrak{p}^e \subseteq Q$. Moreover $Q \cap S = \emptyset$, which gives $Q^c \subseteq \mathfrak{p}$.

Now $\mathfrak{p}^{ec} \subseteq Q^c \subseteq \mathfrak{p}$, and since $\mathfrak{p}^{ec} = \mathfrak{p}$, we get $\mathfrak{p} = Q^c$. □

Proposition 1.68. Let $A \hookrightarrow B$ be rings and C the integral closure of A in B. Let $I \subseteq A$ be an ideal. Then the integral closure of I in B is $\sqrt{IC} \subseteq C$.

Proof. Suppose x is integral over I in B:

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0, \qquad a_i \in I.$$

Then $x \in C$ (a fortiori integral over A), and

$$x^n = -(a_1 x^{n-1} + \cdots + a_n) \in IC,$$

so $x \in \sqrt{IC}$.

Conversely, let $x \in \sqrt{IC}$, so $x^n \in IC$ for some $n > 0$:

$$x^n = a_1 x_1 + \cdots + a_m x_m, \qquad a_i \in I,\ x_i \in C.$$

Each $x_i \in C$ is integral over A, so $M := A[x_1, \ldots, x_m]$ is a finitely generated A-module, and $x^n M \subseteq IM$.

Consider the endomorphism $\mu_{x^n} : M \to M$ given by multiplication by x^n; since $x^n M \subseteq IM$, the determinant-trick argument underlying Nakayama's Lemma (Lecture 6) yields a monic relation

$$(x^n)^k + c_{k-1}(x^n)^{k-1} + \cdots + c_0 = 0, \qquad c_i \in I^i,$$

as endomorphisms of M. Evaluating at $1 \in M$ (since $1 \in A \subseteq M$) gives this same relation as an identity in B, exhibiting x as integral over I. □

Proposition 1.69. Let $A \hookrightarrow B$ be integral domains, $I \subseteq A$ an ideal, and suppose A is integrally closed with $\mathbb{Q}(A) = K$. Let $x \in B$ be integral over I. Then x is algebraic over K, and if

$$\min(x, K) = T^n + a_1 T^{n-1} + \cdots + a_n,$$

then $a_i \in \sqrt{I}$ for all i.

Proof. Clearly x is algebraic over K (being integral over $A \subseteq K$). Let L be an algebraic extension of K containing the conjugates $x_1, \ldots, x_n$ of x.

Each x_i is integral over I (as in the proof of the minimal-polynomial theorem from Lecture 7). Using the relations between roots and coefficients, $a_1, \ldots, a_n$ are all integral over I.

Applying the previous proposition with B replaced by K (in which the integral closure of A is A itself, since A is integrally closed), we get $a_i \in \sqrt{I}$ for all i. □

Theorem 1.70: Going Down. Assume:

(a) $A \hookrightarrow B$ is an integral extension;

(b) A, B are integral domains;

(c) A is integrally closed.

Let $\mathfrak{p}_1 \supseteq \mathfrak{p}_2 \supseteq \cdots \supseteq \mathfrak{p}_n$ be a chain in $\operatorname{Spec} A$, and let $Q_1 \supseteq \cdots \supseteq Q_m$ ($m \leq n$) be a chain in $\operatorname{Spec} B$ with $Q_i \cap A = \mathfrak{p}_i$ for all $i \in [m]$. Then the chain can be extended:

$$Q_1 \supseteq \cdots \supseteq Q_m \supseteq Q_{m+1} \supseteq \cdots \supseteq Q_n$$

with $Q_i \in \operatorname{Spec} B$ and $Q_i \cap A = \mathfrak{p}_i$ for all $i \in [n]$.

Proof. It suffices to treat $m = 1, n = 2$: we need $Q_2 \in \operatorname{Spec} B$ with $Q_1 \supseteq Q_2$ and $Q_2 \cap A = \mathfrak{p}_2$.

The prime ideals of B_{Q_1} are in bijective correspondence with the primes of B contained in Q_1, via

$$A \hookrightarrow B \hookrightarrow B_{Q_1}.$$

It suffices to show $\mathfrak{p}_2 B_{Q_1} \cap A = \mathfrak{p}_2$. The containment $\mathfrak{p}_2 \subseteq \mathfrak{p}_2 B_{Q_1} \cap A$ is clear; we show the reverse.

Let $x \in \mathfrak{p}_2 B_{Q_1} \cap A$, so $x = y/s$ with $y \in \mathfrak{p}_2 B$, $s \in B \setminus Q_1$. Write

$$y = r_1 b_1 + \cdots + r_k b_k, \qquad r_i \in \mathfrak{p}_2,\ b_i \in B.$$

Since B is integral over A, each b_i is integral over A, so y is integral over $\mathfrak{p}_2$ (by the previous propositions).

By the proposition on minimal polynomials (applied with $I = \mathfrak{p}_2$), $\min(y, K)$ (where $K = \mathbb{Q}(A)$) has all coefficients in $\sqrt{\mathfrak{p}_2} = \mathfrak{p}_2$:

$$y^r + u_1 y^{r-1} + \cdots + u_r = 0, \qquad u_i \in \mathfrak{p}_2.$$

Since $x = y/s \in A$, we have $s = yx^{-1}$ with $x^{-1} \in K$ (the case $x = 0$ being trivial). Substituting $y = sx$ into the relation above and dividing by x^r,

$$s^r + \frac{u_1}{x} s^{r-1} + \cdots + \frac{u_r}{x^r} = 0,$$

a minimal equation for s over K. Since $s \in B$ is integral over A, applying the proposition again (with $I = A$) gives $u_i/x^i \in \sqrt{A} = A$ for all i.

Let $v_i := u_i/x^i \in A$, so $v_i x^i = u_i \in \mathfrak{p}_2$. FTSOC, suppose $x \notin \mathfrak{p}_2$. Since $\mathfrak{p}_2$ is prime, $x^i \notin \mathfrak{p}_2$, so $v_i \in \mathfrak{p}_2$ for all i. Then

$$s^r = -\left(\frac{u_1}{x} s^{r-1} + \cdots + \frac{u_r}{x^r}\right) \in \mathfrak{p}_2 B \subseteq \mathfrak{p}_1 B \subseteq Q_1,$$

so $s^r \in Q_1$; since Q_1 is prime, $s \in Q_1$ – contradicting $s \in B \setminus Q_1$.

Hence $x \in \mathfrak{p}_2$, completing the proof. □

Tensor Products of Modules

Definition 1.71. Let R be a commutative ring and M, N, P be R-modules. A *tensor product* of M, N is a pair (T, θ), where T is an R-module and $\theta : M \times N \to T$ is bilinear, such that for every bilinear map $f : M \times N \to P$, there is a unique linear map $h : T \to P$ with

$$\begin{tikzcd} M \times N \arrow[r, "f"] \arrow[d, "\theta"'] & P \\ T \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

i.e. $h \circ \theta = f$.

Exercise 1.72. The tensor product, if it exists, is unique up to isomorphism.

Remark 1.73. $\mathrm{Im}(\theta)$ generates T.

Proof. Let T' be the submodule of T generated by $\mathrm{Im}(\theta)$, with inclusion $i : T' \hookrightarrow T$. Since $\mathrm{Im}(\theta) \subseteq T'$, θ corestricts to a bilinear map $\theta' : M \times N \to T'$. By the universal property of (T, θ) applied to the bilinear map θ' (viewed as landing in T via $i \circ \theta' = \theta$), there is a unique $h : T \to T'$ with $h \circ \theta = \theta'$; composing with i, $(i \circ h) \circ \theta = i \circ \theta' = \theta$.

But id_T also satisfies $\mathrm{id}_T \circ \theta = \theta$; by the uniqueness clause of the universal property, $i \circ h = \mathrm{id}_T$. Hence i is surjective, and since i is also injective (being an inclusion), $T' = T$. □

Construction 1.74. Let (F, ι) be the free R-module on the set $M \times N$. Let H be the submodule of F generated by all elements of the form

(a) $\iota(m_1 + m_2, n) - \iota(m_1, n) - \iota(m_2, n)$,

(b) $\iota(m, n_1 + n_2) - \iota(m, n_1) - \iota(m, n_2)$,

(c) $\iota(rm, n) - r\,\iota(m, n)$,

for $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, $r \in R$. Write $M \otimes_R N := F/H$, and let $\pi : F \to F/H$ be the natural projection.

$\pi \circ \iota$ is bilinear. For instance,

$$(\pi \circ \iota)(m_1 + m_2, n) = \iota(m_1 + m_2, n) + H = \big(\iota(m_1, n) + \iota(m_2, n)\big) + H = (\pi \circ \iota)(m_1, n) + (\pi \circ \iota)(m_2, n),$$

using relation (a); the other bilinearity identities follow similarly from (b) and (c). □

Write $\otimes_R := \pi \circ \iota : M \times N \to M \otimes_R N$, so that $\otimes_R(m, n) =: m \otimes n$ are the *simple tensors*.

$(M \otimes_R N, \otimes_R)$ *satisfies the universal property.* Let P be an R-module and $g : M \times N \to P$ bilinear. Since F is free on $M \times N$, there is a unique morphism $j : F \to P$ such that $j \circ \iota = g$.

$H \subseteq \ker j$**.** For instance,

$$j\big(\iota(m_1 + m_2, n) - \iota(m_1, n) - \iota(m_2, n)\big) = g(m_1 + m_2, n) - g(m_1, n) - g(m_2, n) = 0,$$

using bilinearity of g; similarly for the other generators of H.

Hence j induces a morphism $\bar{k} : M \otimes_R N \to P$ with $\bar{k} \circ \pi = j$, i.e.

$$\bar{k} \circ \otimes_R = \bar{k} \circ (\pi \circ \iota) = (\bar{k} \circ \pi) \circ \iota = j \circ \iota = g.$$

Uniqueness. Let $k : M \otimes_R N \to P$ be any morphism with $k \circ \otimes_R = g$, i.e. $k \circ \pi \circ \iota = g = \bar{k} \circ \pi \circ \iota$. By the uniqueness property of the free module F, $k \circ \pi = \bar{k} \circ \pi$; since π is onto, $k = \bar{k}$. $\square$

CHAPTER 2

PROBLEM DISCUSSIONS

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